CHAPTER

02

Basic Concepts of Logic

Logic is the science that **evaluates arguments**. In this chapter you learn what an argument is and what makes it different from passages that only *look* like arguments. You then learn the two great families of arguments, **deductive** and **inductive**, and how to judge them: **valid, sound, strong, cogent**. This is the most tested chapter of the course, so read every exam trap carefully.

LESSONS IN THIS CHAPTER

2.1 Arguments, Premises and Conclusions

2.2 Techniques of Recognizing Arguments

2.3 Types of Arguments: Deduction and Induction

2.4 Evaluating Arguments

Freshman A+ · Logic & Critical Thinking (PHIL 1011)

CHAPTER 2 · LESSON 2.1

Arguments, Premises and Conclusions

AFTER THIS LESSON YOU CAN

- Define logic and state its benefits
- Tell a statement from other sentences
- Identify premises and conclusions
- Use premise and conclusion indicator words

2.1.1 What is logic?

The word **logic** comes from the Greek word **logos**, which means *sentence, discourse, reason, truth and rule*. The module gives several definitions. All of them say the same thing in different words:

- Logic is **a science that evaluates arguments**.
- Logic is the study of **methods for evaluating whether the premises of arguments adequately support** their conclusions.
- Logic develops **methods and principles** we can use as **criteria for evaluating the arguments of others** and as **guides for constructing our own**.
- Logic is the attempt to **codify the rules of rational thought**. Logicians explore the structure of arguments that preserve truth or allow the optimal extraction of knowledge from evidence.
- Logic is one of the **primary tools philosophers use**: its precision helps them cope with subtle problems and misleading everyday language.

THE ONE-LINE ANSWER

Logic = **the science that evaluates arguments** (the study of the methods and principles of correct reasoning). Its **primary subject matter is the argument.**

Remember from Chapter 1 that logic is both a **field of study** (a branch of philosophy) and an **instrument** (a tool we use to build our own arguments and evaluate others').

Why study logic?

> *"Logic sharpens and refines our natural gifts to think, reason and argue."*
> — C. S. Layman

Studying logic helps us to:

1. **construct sound, fallacy-free arguments** of our own and evaluate the arguments of others;
2. defend against **prejudiced and uncivilized attitudes** that threaten a civilized, democratic society;
3. **distinguish good arguments from bad** ones;
4. identify **common logical errors** in reasoning;
5. identify confusions caused by the **misuse of language**;
6. expose **ill-conceived policies**, and tell the rational from the irrational.

H THE THREE CLASSICAL LAWS OF THOUGHT (OFTEN ASKED IN EXAMS)

- **Law of identity:** *A is A*. Everything is identical to itself.
- **Law of non-contradiction:** nothing can be *both A and not-A* at the same time and in the same respect.
- **Law of excluded middle:** everything *either is or is not*; a statement is either true or false, with no third option.

2.1.2 What is an argument?

In everyday life "argument" can mean a quarrel. In logic it has a **very specific, technical meaning**:

D ARGUMENT

A **group of statements**, one or more of which (the **premises**) are claimed to provide support for, or reasons to believe, one of the others (the **conclusion**).

Look at the two parts of this definition.

First, an argument is made of statements. A **statement** is a **declarative sentence that is either true or false**. True and false are its two possible **truth values**.

★ THESE ARE STATEMENTS

- "Mekelle is the capital city of Tigray." *(true)*
- "Ethiopia was colonized by Germany." *(false, but still a statement)*

! THESE ARE NOT STATEMENTS

- "Would you close the window?" *(question)*
- "Let us study together." *(proposal)*
- "Right on!" *(exclamation)*
- "I suggest that you read philosophy texts." *(suggestion)*
- "Give me your ID card, now!" *(command)*

None of the sentences on the right can be true or false, so **none of them can be part of an argument.**

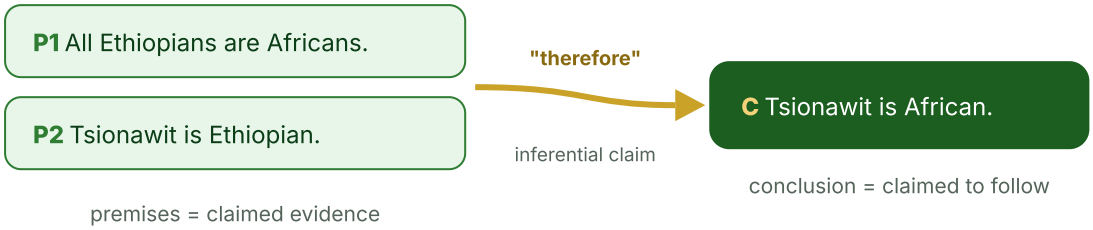
I STATEMENT VS PROPOSITION

Logicians often use the two words interchangeably. Strictly, a **proposition** is the **meaning or information content** of a statement.

Second, the statements are divided into premises and a conclusion. An argument has **at least one premise** and **one and only one conclusion**.

Part	What it is	Ask yourself
Premise	The statement(s) that give the **reason or evidence**: the *claimed evidence*	"What is offered as support?"

Part	What it is	Ask yourself
Conclusion	The statement **claimed to follow** from the evidence: the claim the argument tries to establish	"What is the arguer trying to prove?"



Every argument has the same skeleton: evidence → (claimed) support → conclusion

Good arguments and bad arguments

All arguments fall into two groups: those whose premises **really do** support the conclusion (**good**, well-supported) and those whose premises **do not**, even though they claim to (**bad**, poorly supported).

E GOOD VS BAD, BUT BOTH ARE ARGUMENTS

Good: All Ethiopians are Africans. Tsionawit is Ethiopian. Therefore, Tsionawit is African.
Bad: Some Africans are black. Zelalem is African. Therefore, Zelalem is black.
The second gives weak support, **but it is still an argument**, because it *claims* that the conclusion follows.

2.1.3 How to find the conclusion: indicator words

★ CONCLUSION INDICATORS

therefore · wherefore · thus · consequently · we may infer · accordingly · we may conclude · it must be that · for this reason · so · entails that · hence · it follows that · implies that · as a result · whence · it shows that

E PREMISE INDICATORS

since · as indicated by · because · for · in that · may be inferred from · as · given that · seeing that · for the reason that · inasmuch as · owing to

Rule: the statement that comes **after a conclusion indicator** is usually the conclusion. The statement that comes **after a premise indicator** is usually a premise. The remaining statements are identified by elimination.

! "FOR THIS REASON" VS "FOR THE REASON THAT"

- **"For this reason"** comes *after* the premise and *before* the conclusion. The statement before it is the premise; the statement after it is the conclusion. So it acts as **both** a premise and a conclusion indicator.
- **"For the reason that"** is an ordinary **premise** indicator.

Example: *"The dam provides electricity. For this reason, the region's factories can grow."* The conclusion is the sentence after the phrase.

One indicator can introduce more than one premise. In "Tsionawit is a faithful wife, *for* Ethiopian women are faithful wives *and* Tsionawit is an Ethiopian," the word *for* introduces **two** premises.

No indicator at all? Then ask:

1. What single statement is claimed (implicitly) to follow from the others?
2. What is the arguer trying to prove?
3. What is the **main point** of the passage?

E AN ARGUMENT WITH NO INDICATOR WORDS

"Our country should increase the quality and quantity of its military. Ethnic conflicts have recently intensified; border conflicts are escalating; international terrorist activities are increasing."
C: The country should increase its military. **P1–P3:** the three reasons. *(When there are no indicators, the conclusion is often the first statement.)*

Not every sentence is a premise or the conclusion. Leave out passing comments. In *"Socialized medicine is not recommended because it would reduce the quality of care. In addition, it might bankrupt the treasury. This is the whole case against socialized medicine in a nutshell."*, the last sentence is only a comment about the argument.

I INFERENCE

In its **narrow** sense, *inference* means the **reasoning process** expressed by an argument. In its **broad** sense it means the argument itself. The module uses the narrow sense: the inferential link between premises and conclusion.

Exam Practice

12 QUESTIONS · LESSON 2.1

1 Jigjiga University · Mid 2016

"Logic is the attempt to codify the rules of rational thought. Logicians explore the structure of arguments that preserve truth or allow the optimal extraction of knowledge from evidence."

True / False

2 Jigjiga University · Mid 2014

Any sentence that is either true or false is a statement.

True / False

3 Jigjiga University · Mid 2014

Which of the following is an argument?

A) Ali is angry with me, so he probably won't be willing to help me wash the dishes.
B) Ali washed the dishes, then he went to bed.
C) If Ali is angry, then he won't help me.
D) Ali is tall and Ali is kind.
E) Is Ali angry? What a mess!

4 Jigjiga University · Mid 2014

Which option contains ONLY conclusion indicators?

A) Therefore, since, hence
B) Thus, owing to, accordingly
C) As a result, consequently, for this reason
D) It follows that, seeing that, so
E) Hence, because, we may infer

5 Oda Bultum University · Mid 2017 E.C.

Which of the following indicators can be both a premise indicator and a conclusion indicator?

A) Consequently
B) Since
C) For this reason
D) As a result

6 Jigjiga University · Mid 2014

Which of the following is INCORRECT about an argument?

A) It is a technical term
B) It is a group of statements
C) It does not mean a mere verbal fight
D) It has a general meaning in logic

7 Jigjiga University · Mid 2014

Which one is FALSE about logic?

A) Logic is a branch of philosophy that deals with arguments
B) Logic is an instrument we can use to formulate our own rational arguments
C) Logic helps develop methods and principles for evaluating arguments
D) Logic sharpens and refines our natural gifts to think, reason and argue
E) None of the above

8 Oda Bultum University · Mid 2017 E.C.

The law of identity is one of Aristotle's three laws of logic. Which statement best exemplifies it?

A) A is the same as itself
B) A can be both A and not A at the same time
C) A is always equal to B
D) A can be different from itself

9 Jigjiga University · Final

The rule "Something either is or is not" is called the:

A) Law of identity
B) Law of excluded middle
C) Law of non-contradiction
D) Law of sufficient reason

10 Jigjiga University · Mid 2014

"All machines have a finite working lifetime, and even though that big tree doesn't look like a typical machine, it is really just a biological machine; I believe it will stop someday, too." What is the conclusion?

A) Even though that big tree doesn't look like a typical machine
B) All machines have a finite working lifetime
C) That big tree is a biological machine
D) That big tree will stop (die) someday, too

11 Hurley · Exercise 1.1

Identify the premises and conclusion: "It really does matter if you get enough sleep. We need sleep to think clearly, react quickly, and create memories. Studies show that people who are taught mentally challenging tasks do better after a good night's sleep. Other research suggests that sleep is needed for creative problem solving."

12 Addis Ababa University · Model Exam

"Tsionawit is a faithful wife, for Ethiopian women are faithful wives and Tsionawit is an Ethiopian." How many premises does the indicator "for" introduce?

A) None
B) One
C) Two
D) Three

Answers & Explanations

1. True

This is one of the module's own definitions of logic.

2. True

A statement is a declarative sentence that has a truth value of either true or false.

3. A

Ali is angry with me, so he probably won't be willing to help me wash the dishes.

"So" introduces a conclusion supported by "Ali is angry". B reports events in time order, C is a single conditional statement (not an argument), D is a plain statement, and E contains a question and an exclamation, which are not statements.

4. C

As a result, consequently, for this reason

Each of the other options hides one premise indicator: since (A), owing to (B), seeing that (D) and because (E). The statement after "for this reason" is the conclusion.

5. C

For this reason

The statement before "for this reason" is the premise and the statement after it is the conclusion.

6. D

It has a general meaning in logic

The module says the term "argument" has a very **specific** meaning in logic.

7. E

None of the above

All four statements come straight from the module, so none is false.

8. A

A is the same as itself

The law of identity says "A is A". B breaks the law of non-contradiction.

9. B

Law of excluded middle

Excluded middle: there is no third ("middle") possibility between is and is not. Non-contradiction says something cannot be both A and not-A.

10. D

That big tree will stop (die) someday, too

The conclusion is the claim being proved. B and C are the premises; A is only a passing remark inside a premise. (The original paper printed only option A; the other options are added here.)

11. C = It really does matter if you get enough sleep

The other three sentences are reasons (P1, P2, P3). With no indicator words, ask "what is the arguer trying to prove?" The answer is the first sentence, which is common in passages without indicators.

12. C

Two

A single indicator can introduce more than one premise. "For" goes with both "Ethiopian women are faithful wives" and "Tsionawit is an Ethiopian".

CHAPTER 2 · LESSON 2.2

Techniques of Recognizing Arguments

AFTER THIS LESSON YOU CAN

- State the two conditions an argument must meet
- Distinguish factual and inferential claims
- Recognize the main kinds of non-arguments
- Explain necessary and sufficient conditions

2.2.1 When does a passage contain an argument?

A passage contains an argument if it **purports to prove something**. That requires **two conditions**:

#	Condition	Name	Who checks it?
1	At least one statement must **claim to present evidence or reasons**	**Factual claim**	Often outside logic
2	There must be a **claim that the evidence supports or implies something**: that something *follows*	**Inferential claim**	**Logic's main concern**

★ **THE KEY TEST**

It is **not necessary** that the premises present *true* evidence, or that they *actually* support the conclusion. They only have to **claim** to. A bad argument is still an argument, but a passage with **no inferential claim** is **not** an argument.

The inferential claim is **not the same as the intentions of the arguer**. Intentions are subjective and hidden; the inferential claim is an **objective feature** grounded in the passage's **language or structure**.

Explicit and implicit inferential claims

- **Explicit:** shown by an indicator word. *"Gamachuu is my biological father, because my mother told me so."*
- **Implicit:** there is an inferential relationship but **no indicator word**. *"The genetic modification of food is risky business. Genetic engineering can introduce unintended changes into the DNA of the food-producing organism, and these changes can be toxic to the consumer."*

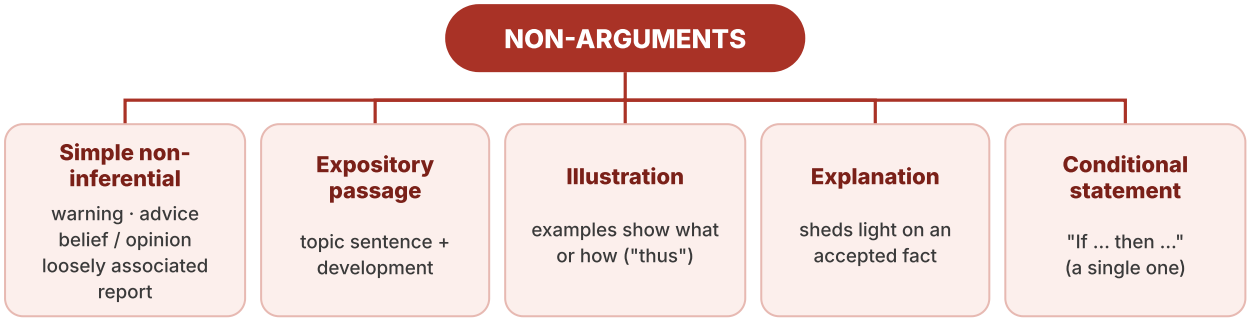
! **AN INDICATOR WORD DOES NOT GUARANTEE AN ARGUMENT**

- *"Since Edison invented the phonograph, there have been many technological developments."* → "since" means **from the time that**, so this is **not** an argument.
- *"Since Edison invented the phonograph, he deserves credit for a major technological development."* → "since" is **logical**, so this **is** an argument.

Tip: mentally insert "therefore" between the statements and see whether the ideas flow.

2.2.2 Recognizing non-arguments

Non-arguments **lack an inferential claim**. They may contain statements that *could* be premises or conclusions, but nothing is claimed to be proved.



The typical kinds of non-arguments in the module

A. Simple non-inferential passages

Type	Meaning	Module example
Warning	Puts someone on guard against danger	"Whatever you promise to tell, never confide political secrets to your wife."
Piece of advice	Recommends a future decision or conduct	"After class hours, I would suggest that you give careful consideration to the subject matter."
Statement of belief or opinion	Expresses what someone believes or thinks	"We believe that our university must develop and produce outstanding students..."
Loosely associated statements	Same general subject, but none is claimed to prove another	Lao-Tzu: "Not to honor men of worth will keep the people from contention; not to value goods that are hard to come by will keep them from theft..."
Report	Statements that convey information about a topic or event	"The Grand Renaissance Dam has opened employment for thousands of Ethiopians..."

REPORTS ABOUT ARGUMENTS

"'The Air Force faces a serious shortage of pilots, because ...,' says an Air Force official." The author only **reports** someone else's argument, so strictly the passage is **not** the author's argument. If you treat it as one, make clear whose argument it is.

B. Expository passages

A **topic sentence** followed by sentences that **develop** it. If the aim is only to **expand or elaborate** the topic sentence, there is **no argument**. But if the following sentences also try to **prove** it, the passage **is** an argument.

HOW TO DECIDE (EXPOSITORY PASSAGES AND ILLUSTRATIONS)

Ask: does the passage only show *what something means* or *how it is done*, or does it also try to **prove** something? In borderline cases, ask whether the claim is one **everyone already accepts**. Authors rarely try to prove what everyone accepts, so if the claim is accepted the passage is probably **not** an argument. If many people would doubt it, it probably **is** one.

C. Illustrations

One or more **examples** intended to show **what something means or how it is done**. They often contain "thus", so they are easily confused with arguments.

ILLUSTRATION (NOT AN ARGUMENT)

"Chemical elements, as well as compounds, can be represented by molecular formulas. **Thus**, oxygen is represented by 'O2', water by 'H2O', and sodium chloride by 'NaCl'."
"Thus" here shows **how** something is done.

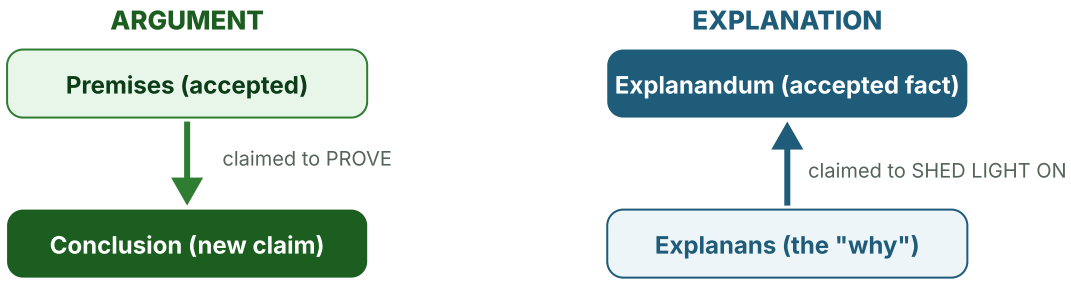
ARGUMENT FROM EXAMPLE (AN ARGUMENT)

"Although most forms of cancer, if untreated, can cause death, not all cancers are life-threatening. **For example**, basal cell carcinoma... almost never results in death."
The example is used to **prove** "not all cancers are life-threatening".

D. Explanations

An **explanation** tries to **shed light on** some event or phenomenon that is **usually accepted as a fact**: it tells **why** something is so.

- **Explanandum** = the statement that describes the event **to be explained**.
- **Explanans** = the statement(s) that **do the explaining**.



An argument moves from accepted premises to a new claim; an explanation starts from an accepted fact and asks why

COWS AND GRASS

"Cows digest grass while humans cannot, **because** their digestive systems contain an enzyme not found in humans."
Explanandum: Cows digest grass while humans cannot (everyone accepts this). **Explanans:** their digestive systems contain an enzyme... This is **not trying to prove** that cows digest grass; it explains **why**.

★ TEST FOR EXPLANATIONS

Find the statement that is either the explanandum or the conclusion (usually the one **before "because"**). If it describes an **accepted matter of fact**, and the rest shed light on it, the passage is an **explanation**. Some passages can be **both**: "Women become intoxicated by drinking a smaller amount of alcohol than men because men metabolize part of the alcohol before it reaches the bloodstream."

E. Conditional statements

A **conditional statement** is an **"if … then …"** statement. *"If you study hard, then you will score an A."*

- **Antecedent** = the part after **"if"** (*you study hard*).
- **Consequent** = the part after **"then"** (*you will score an A*).
- If "then" is left out, the order may be reversed: *"You will score an A if you study hard."* The antecedent is still the part after "if".
- A meaningful link is **not required**: "If Getaneh Kebede is a singer, then Hawassa is in Mekelle" is still a conditional.

A single conditional is **not an argument**, because it does **not assert** that the antecedent or the consequent is true. It only says *if* the antecedent is true, *then* so is the consequent.

★ THE THREE RULES: CONDITIONALS AND ARGUMENTS

1. **A single conditional statement is not an argument.**
2. **A conditional statement may serve as the premise, the conclusion, or both** of an argument.
3. **The inferential content of a conditional statement may be re-expressed to form an argument.** "If destroying a political competitor gives you joy, then you have a low sense of morality" → "Destroying a political competitor gives you joy. **Therefore**, you have a low sense of morality."

2.2.3 Necessary and sufficient conditions

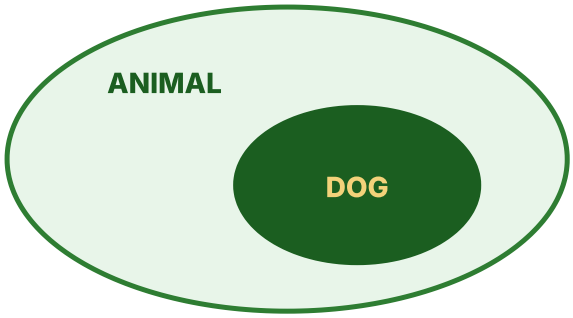
Conditional statements express **necessary and sufficient conditions**.

D SUFFICIENT CONDITION

A is sufficient for B when the occurrence of A is **all that is needed** for B. *Being a dog is sufficient for being an animal.*
→ **If A, then B.** ("If X is a dog, then X is an animal.")

D NECESSARY CONDITION

B is necessary for A when A **cannot occur without** B. *Being an animal is necessary for being a dog.*
→ **If not B, then not A.** ("If X is not an animal, then X is not a dog.")



Inside DOG → surely inside ANIMAL
dog is SUFFICIENT for animal

Outside ANIMAL → surely no DOG
animal is NECESSARY for dog

Inside ANIMAL ≠ surely a dog
(it could be a cat or a mouse)

The module's closed-box example: the small circle is sufficient for the big one; the big one is necessary for the small one

M ANTECEDENT = SUFFICIENT, CONSEQUENT = NECESSARY

In **"If A, then B"**: **A** (antecedent) is **sufficient** for B, and **B** (consequent) is **necessary** for A. Think "**S**mall circle = **S**ufficient; big circle = necessary."

E TEST YOURSELF

- "Being in Kenya" is **sufficient** for "being in Africa". "Being in Africa" is **necessary** for "being in Kenya".
- "Having a racket" is **necessary** for playing tennis. (Hurley)
- "Burning leaves" is **sufficient** for producing smoke. (Hurley)
- "If the car is running, there is gas in the tank": running is sufficient for gas; gas is necessary for running.

2.2.4 Summary checklist: is it an argument?

Look for three things: **(1)** indicator words, **(2)** an inferential relationship between the statements, and **(3)** the typical kinds of non-arguments. An indicator word alone does not guarantee an argument. In arguments **without** indicators, the conclusion is often the **first** statement.

Exam Practice

19 QUESTIONS · LESSON 2.2

1 Jigjiga University · Mid 2014

A single conditional statement is not an argument.

True / False

2 Salale University · Mid 2015

If a passage contains an explicit inferential claim indicated by "therefore", it is an argument, whether or not the premises actually support the conclusion.

True / False

3 Salale University · Mid 2015

A passage with two or more statements but no inferential claim is still an argument if the statements are on the same topic.

True / False

4 Oda Bultum University · Mid 2017 E.C.

Being in Africa is a sufficient condition for being in Kenya.

True / False

5 Salale University · Mid 2015

A necessary condition for something to be an argument is that:

A) all premises must be true
B) the conclusion must be true
C) there must be an inferential claim
D) it must be deductive

6 Salale University · Mid 2015

Which passage is a non-argument that could easily be mistaken for an argument?

A) All metals expand when heated; iron is a metal; so iron expands when heated
B) The ground is wet because it rained last night
C) Since the bridge is closed, we must take the long road
D) If it rains, the match will be cancelled

7 Jigjiga University · Mid 2014

"We believe that our country planned to increase the national economy in order to reduce poverty." This is a:

A) Warning
B) Piece of advice
C) Loosely associated statements
D) Statement of belief (opinion)

8 Jigjiga University · Mid 2014

"Sorrow and disbelief descended on Japan after Shinzo Abe, the former prime minister, was shot dead while giving a campaign speech on Friday morning." This passage is:

A) A deductive argument
B) An inductive argument
C) A report (not an argument)
D) An illustration

9 Jigjiga University · Mid 2014

"If the car is running, there's gas in the tank." This shows that:

A) Gas in the tank is necessary for the car to run
B) The car running is sufficient for there being gas
C) The antecedent is sufficient for the consequent
D) The consequent is necessary for the antecedent
E) All of the above

10 Jigjiga University · 2012

"If leaves burn, then smoke is produced." What is the burning of leaves for the production of smoke?

A) Sufficient condition
B) Necessary condition
C) Antecedent only
D) Consequent

11 Jigjiga University · 2012

Which of the following groups are ALL arguments?

A) Illustrations, explanations, generalizations
B) Warnings, conditional statements, statements of belief
C) Expository passages, reports, pieces of advice
D) Generalizations, categorical syllogisms, arguments from signs

12 Salale University · Mid 2015

Which passage contains an IMPLICIT inferential claim?

A) The match was cancelled because the stadium flooded.
B) The genetic modification of food is risky business. Genetic engineering can introduce unintended changes into DNA, and these changes can be toxic to the consumer.
C) If the dam is completed, electricity will be cheaper.
D) Close the door when you leave.

13 Oda Bultum University · Mid 2017 E.C.

Which correctly describes necessary and sufficient conditions?

A) A necessary condition guarantees the event; a sufficient condition must always be present
B) A necessary condition must always be present for the event; a sufficient condition guarantees the event
C) A necessary condition guarantees the event; a sufficient condition may be absent
D) Both must always be present

14 Oda Bultum University · Mid 2017 E.C.

Which statement accurately summarizes the relationship between conditional statements and arguments?

A) A conditional statement may serve as either the premise or the conclusion (or both) of an argument
B) A single conditional statement is always an argument
C) Conditional statements can only serve as premises
D) The inferential content of a conditional cannot be re-expressed as an argument

15 Jigjiga University · Mid 2014

Fill in the blanks: (1) **__ purports to shed light on an event accepted as fact. (2) _ *shows by examples what something means or how it is done. (3)*** ___ is a topic sentence followed by sentences that develop it.

16 Hurley · Exercise 1.2

"The turkey vulture is called by that name because its red featherless head resembles the head of a wild turkey." This passage is:

A) An argument
B) An explanation
C) A report
D) A warning

17 Hurley · Exercise 1.2

"A mammal is a vertebrate animal that nurses its offspring. Thus, cats and dogs are mammals, as are sheep, monkeys, rabbits, and bears." This passage is:

A) An argument from example
B) An illustration
C) An expository passage that proves its topic
D) A conditional statement

18 Hurley · Exercise 1.2

"A person never becomes truly self-reliant. Even though he deals effectively with things, he is necessarily dependent upon those who have taught him to do so." (B. F. Skinner) This passage is:

A) A loosely associated set of statements
B) An argument with the conclusion "A person never becomes truly self-reliant"
C) An explanation of an accepted fact
D) A report

19 Hurley · Exercise 1.2

Fill in the blank with "necessary" or "sufficient": "Having a racket is a _____ condition for playing tennis."

A) necessary
B) sufficient

Answers & Explanations

1. True

This is the first rule in the module. A conditional asserts neither its antecedent nor its consequent, so it makes no inferential claim.

2. True

A passage is an argument when it **claims** that something follows; actual support is not required. (Careful: the word "therefore" alone is not enough. The statement is true because it says there really is an inferential claim.)

3. False

These are loosely associated statements. They share a subject, but none is claimed to prove another.

4. False

Being in Africa is **necessary** for being in Kenya. Being in Kenya is sufficient for being in Africa.

5. C

there must be an inferential claim

Arguments can have false premises or a false conclusion, and inductive arguments are arguments too. What every argument needs is the claim that something follows.

6. B

The ground is wet because it rained last night

B is an **explanation**: everyone accepts that the ground is wet, and the "because" clause explains why. Explanations are often mistaken for arguments because they contain "because". D is a non-argument too, but it is not usually confused with one.

7. D

Statement of belief (opinion)

It expresses what someone believes, with no claim that the belief is supported by evidence. It is like the module's "We believe that our university must...".

8. C

A report (not an argument)

It conveys information about an event; nothing is claimed to be proved, so there is no inferential claim. (The original paper offered only A and B, which were both wrong.)

9. E

All of the above

In "If A, then B", A is sufficient for B and B is necessary for A. All four options say this in different words.

10. A

Sufficient condition

The antecedent is sufficient for the consequent. Burning leaves is not **necessary**, because smoke can come from other things.

11. D

Generalizations, categorical syllogisms, arguments from signs

Inductive generalization, the deductive categorical syllogism and the inductive argument from signs are all kinds of argument. Every other option contains at least one non-argument.

12. B

The genetic modification of food is risky business. Genetic engineering can introduce unintended changes into DNA, and these changes can be toxic to the consumer.

B has an inferential link but no indicator word; it is the module's own example. A uses "because" (and is an explanation), C is a single conditional, and D is a command.

13. B

A necessary condition must always be present for the event; a sufficient condition guarantees the event

Without a necessary condition the event cannot occur. A sufficient condition is all that is needed for the event to occur.

14. A

A conditional statement may serve as either the premise or the conclusion (or both) of an argument

This is rule 2. B contradicts rule 1, C is too narrow, and D contradicts rule 3.

15. Explanation · Illustration · Expository passage

These are the module's definitions of three kinds of non-arguments.

16. B

An explanation

It is accepted that the bird has this name; the "because" clause explains **why**. (Hurley's answer: nonargument, explanation.)

17. B

An illustration

The examples show **what** "mammal" covers; nothing doubtful is being proved, even though "thus" appears. (Hurley: nonargument, illustration.)

18. B

An argument with the conclusion "A person never becomes truly self-reliant"

The first sentence is a claim many people would doubt, and the second gives a reason for it. With no indicator words, the conclusion is the first statement.

19. A

necessary

You cannot play tennis without a racket, but having a racket does not guarantee you are playing. As a conditional: "If a person plays tennis, then he or she has a racket."

CHAPTER 2 · LESSON 2.3

Types of Arguments: Deduction and Induction

AFTER THIS LESSON YOU CAN

- Define deductive and inductive arguments
- Use the three factors to tell them apart
- Name the deductive and inductive argument forms
- Reject the "general vs particular" myth

2.3.1 Two kinds of inferential claim

Every argument claims that its premises support its conclusion. The difference between the two great families is **how strongly** the conclusion is claimed to follow.

D DEDUCTIVE ARGUMENT

The premises are claimed to support the conclusion so that **it is impossible for the premises to be true and the conclusion false.** The conclusion follows **necessarily**, with **certainty**.

All philosophers are critical thinkers. Socrates is a philosopher. Therefore, Socrates is a critical thinker.

D INDUCTIVE ARGUMENT

The premises are claimed to support the conclusion so that **it is improbable for the premises to be true and the conclusion false.** The conclusion follows only **probably**.

Most African leaders are black. Mandela was an African leader. Therefore, probably Mandela was black.

I IN SIMPLE WORDS

Deduction promises **"must"**; **induction** promises only **"probably"**. In induction the conclusion "goes beyond" the premises: it moves from something familiar to something less known.

2.3.2 The three factors for deciding

Most arguments do not say how strongly the conclusion follows, so we look at **three objective features**:

Factor 1 · Special indicator words

Suggests deductive	Suggests inductive
certainly · necessarily · absolutely · definitely	probable · improbable · plausible · implausible · likely · unlikely · reasonable to conclude

! INDICATOR WORDS CAN MISLEAD

- **"It must be the case that"** is **ambiguous**: "must" can indicate probability *or* necessity.
- People say "it **certainly** follows" just for rhetorical effect, and some claim to "deduce" what is really an induction.
- If an indicator word **conflicts** with the other two factors, **ignore it**.

Factor 2 · The actual strength of the inferential link

If the conclusion **actually follows with strict necessity**, the argument is **deductive**. If it does not follow necessarily but **does follow probably**, it is usually best read as **inductive**.

E COMPARE

All Ethiopian people love their country. Debebe is an Ethiopian. Therefore, Debebe loves his country. → **deductive**.
The majority of Ethiopian people are poor. Alamudin is an Ethiopian. Therefore, Alamudin is poor. → **inductive**.

Factor 3 · The character or form of the argument

When there are no indicator words and the conclusion follows *neither* necessarily *nor* probably, look at the **form**.

Deductive forms	Inductive forms
Argument based on mathematics: depends on arithmetic or geometric computation or measurement (2 oranges + 3 bananas = 5 fruits)	**Prediction**: from present or past events to the **future** (clouds → rain within 24 hours)
Argument from definition: depends on the definition of a word ("Kebede is a physician; therefore he is a doctor")	**Argument from analogy**: from a **similarity** between two things (Computer A is fast; B is similar; so B is fast)
Categorical syllogism: two premises and a conclusion, each beginning with **all, no, some**	**Inductive generalization**: from a **sample** to the **whole group** (3 out of 4 in one prison → 3/4 of all prisoners)
Hypothetical syllogism: a **conditional** ("if ... then") as one or both premises	**Argument from authority**: rests on an **expert or witness** (eyewitness testimony; "the professor said so")
Disjunctive syllogism: an **"either ... or"** premise ("Rewina is either Ethiopian or Eritrean; she is not Eritrean; so she is Ethiopian")	**Argument from signs**: from a **sign** to what it symbolizes (a "No Parking" sign)
	Causal inference: from **cause to effect** (water left in the freezer → frozen) or **effect to cause** (dry, tough chicken → overcooked)

M **DEDUCTIVE: "MATH, DEFINITIONS & 3 SYLLOGISMS"; INDUCTIVE: "PAGASC"**

Prediction · **A**nalogy · **G**eneralization · **A**uthority · **S**igns · **C**ausal inference.

! **SPECIAL CASES STUDENTS MISS**

- **Statistics:** arguments that depend on **statistics** are usually **inductive**, even though mathematics is deductive.
- **Geometry** arguments are **always deductive**. Don't mistake "triangle A is congruent with triangle B, so B has the same angle" for an analogy.
- **Science:** arguments aimed at **discovering a law of nature** are usually **inductive**; arguments **applying** known laws to cases are often **deductive** (with reservations).
- The inductive forms **overlap**: many causal inferences are also predictions.

2.3.3 The "general vs particular" myth

An old tradition (going back to Aristotle) says *induction goes from particular to general and deduction from general to particular*. **This is wrong.** Both kinds can go in **any direction**.

E **DEDUCTIVE: PARTICULAR → GENERAL**

Three is a prime number. Five is a prime number. Seven is a prime number. Therefore, **all** odd numbers between two and eight are prime numbers.

E **INDUCTIVE: GENERAL → PARTICULAR**

All emeralds previously found have been green. Therefore, the **next** emerald to be found will be green.

Exam Practice

21 QUESTIONS · LESSON 2.3

1 Jigjiga University · Mid 2014

A geometrical proof is an example of an inductive argument.

True / False

2 Jigjiga University · Mid 2014

An argument that proceeds from knowledge of a cause to knowledge of an effect is an inductive argument.

True / False

3 Jigjiga University · Mid 2014

Deductive arguments always proceed from the general to the particular.

True / False

4 Salale University · Mid 2015

The distinction between deductive and inductive arguments is based solely on indicator words such as "necessarily" or "probably".

True / False

5 Oda Bultum University · Mid 2017 E.C.

In a deductive argument, the conclusion is not supposed to contain more information than the premises.

True / False

6 Jigjiga University · Mid 2014

"Since November, when the inflationary spiral ended, state taxes have been high. State farm subsidies will therefore continue to rise." This is:

A) Deductive argument from authority
B) Deductive argument from definition
C) Inductive argument based on prediction
D) Causal inference only
E) None

7 Jigjiga University · Mid 2014

"All sharks are birds. All birds are politicians. Therefore, all sharks are politicians." What kind of deductive argument is this?

A) Categorical syllogism
B) Hypothetical syllogism
C) Disjunctive syllogism
D) Argument based on mathematics
E) Argument from definition

8 Jigjiga University · Mid 2014

Which word indicates a deductive argument?

A) Certainly
B) Probable
C) Plausible

9 Jigjiga University · 2012

"If large dams are constructed on the Tekeze River, then the quantity of water arriving in Sudan reduces. Large dams are constructed. Therefore, the quantity of water arriving in Sudan reduces." This is a:

A) Categorical syllogism
B) Disjunctive syllogism
C) Hypothetical syllogism
D) Inductive argument

10 Jigjiga University · 2012

"Since $2x + y = 20$ and $x = 4$, it follows that $y = 12$." This argument is:

A) Deductively valid
B) Deductively invalid
C) Inductively strong
D) Inductively weak

11 Jigjiga University · 2012

A judge decides a case based on testimony from eyewitnesses under oath. This is an:

A) Argument from definition
B) Argument from authority
C) Argument from analogy
D) Argument from signs

12 Jigjiga University · 2012

"Tigers are striped. Zebras are also striped. Tigers are carnivores. It implies that zebras are also carnivores." This argument is:

A) Deductive, valid
B) Inductive, weak
C) Deductive, invalid
D) Inductive, strong

13 Salale University · Mid 2015

"Since all bachelors are unmarried men, and all unmarried men are adult males, all bachelors are adult males." This is:

A) Inductive, strong
B) A categorical syllogism; valid
C) A hypothetical syllogism
D) A disjunctive syllogism

14 Salale University · Mid 2015

Which statement about inductive arguments is correct?

A) They always go from particular to general
B) Their conclusions are certain if the premises are true
C) A strong argument with a false premise is still cogent
D) They can go particular → general, general → particular, or particular → particular

15 Salale University · Mid 2015

"The majority of students in this class have smartphones. Therefore, the next student I meet will probably have a smartphone." This is:

A) Deductive, valid
B) Inductive, strong
C) Inductive, weak
D) Deductive, invalid

16 Oda Bultum University · Mid 2017 E.C.

Which one of the following is the valid structure of a hypothetical syllogism?

A) If P, then Q. If Q, then R. Therefore, if P, then R.
B) All A are B. All B are C. Therefore, all A are C.
C) Either A or B. Not A. Therefore, B.
D) If P, then Q. If Q, then R. Therefore, if R, then Q.

17 Oda Bultum University · Mid 2017 E.C.

What is the primary focus of an argument from sign?

A) Drawing conclusions based on observable indicators
B) Using emotional appeals
C) Ignoring the relevance of evidence
D) Deductive reasoning

18 Hurley · Exercise 1.3

"Because triangle A is congruent with triangle B, and triangle A is isosceles, it follows that triangle B is isosceles." This argument is:

A) Inductive, argument from analogy
B) Deductive, argument based on mathematics (geometry)
C) Inductive, generalization
D) Inductive, causal inference

19 Hurley · Exercise 1.3

"World-renowned physicist Stephen Hawking says that the condition of the universe at the Big Bang was more highly ordered than it is today. In view of Hawking's stature, we should conclude that this description is correct." This argument is:

A) Deductive, from definition
B) Inductive, from authority
C) Inductive, prediction
D) Deductive, categorical syllogism

20 Hurley · Exercise 1.3

"Though someone had fetched out a horse, the dog in the stables had not barked enough to arouse the two lads in the loft. Obviously the midnight visitor was someone whom the dog knew well." (Sherlock Holmes) This argument is:

A) Deductive, despite the word "obviously"
B) Inductive, a causal inference
C) Deductive, from definition
D) Not an argument

21 Jigjiga University · Mid 2016

"In the recent election, the PP party is leading in almost all regions according to reliable sources; therefore, it's more likely they will be the winners." What kind of argument is this?

Answers & Explanations

1. False

Arguments in geometry are always deductive.

2. True

It is a causal inference. Specific instances of cause and effect can never be known with absolute certainty, so it is inductive.

3. False

The module rejects this tradition. "Three, five and seven are prime; therefore all odd numbers between two and eight are prime" is deductive but goes from particular to general.

4. False

There are **three** factors: indicator words, the actual strength of the link, and the form of the argument.

5. True

In induction the conclusion "goes beyond" the premises; in deduction it follows from them with necessity.

6. C

Inductive argument based on prediction

The premises describe the past and present; the conclusion is about the **future** ("will continue"). Predictions about the future are inductive.

7. A

Categorical syllogism

It has exactly two premises and one conclusion, each beginning with "all". (It is valid but unsound, because the premises are false.)

8. A

Certainly

"Certainly", "necessarily", "absolutely" and "definitely" suggest deduction; "probable" and "plausible" suggest induction.

9. C

Hypothetical syllogism

A hypothetical syllogism has a conditional ("if ... then") statement as one or both premises.

10. A

Deductively valid

It is based on mathematics, so it is deductive. Check: 2(4) + 12 = 20, so the conclusion follows necessarily and the argument is valid.

11. B

Argument from authority

The conclusion rests on statements made by a presumed authority or **witness**. It is inductive, because witnesses can be mistaken or lie.

12. B

Inductive, weak

It is an argument from analogy, which is inductive. Stripes have no causal or systematic link to diet, so the analogy is weak (zebras are actually herbivores).

13. B

A categorical syllogism; valid

Its form is "All A are B; all B are C; so all A are C", which is valid.

14. D

They can go particular → general, general → particular, or particular → particular

Both deductive and inductive arguments can go in any direction. A strong argument with a false premise is **un**cogent.

15. B

Inductive, strong

"Probably" and the majority premise make it inductive, and a majority gives good support, so it is strong.

16. A

If P, then Q. If Q, then R. Therefore, if P, then R.

B is a categorical syllogism, C a disjunctive syllogism, and D is invalid.

17. A

Drawing conclusions based on observable indicators

It reasons from a sign, such as a "No Parking" sign, to the situation the sign symbolizes.

18. B

Deductive, argument based on mathematics (geometry)

Geometry arguments are always deductive. Do not mistake "congruent with" for an analogy.

19. B

Inductive, from authority

The conclusion rests on an expert's statement, and experts can be mistaken.

20. B

Inductive, a causal inference

It reasons from an effect (the silent dog) to a cause (familiarity with the visitor). Ignore the word "obviously", which only adds rhetorical force.

21. **Inductive (prediction), strong**

The conclusion concerns a future event and follows only probably ("more likely"). Strong support from reliable sources makes it strong, and cogent if the premise is true.

CHAPTER 2 · LESSON 2.4

Evaluating Arguments

AFTER THIS LESSON YOU CAN

- Test the inferential claim before the factual claim
- Define valid, invalid and sound
- Define strong, weak and cogent
- Use the truth-value tables for validity and strength

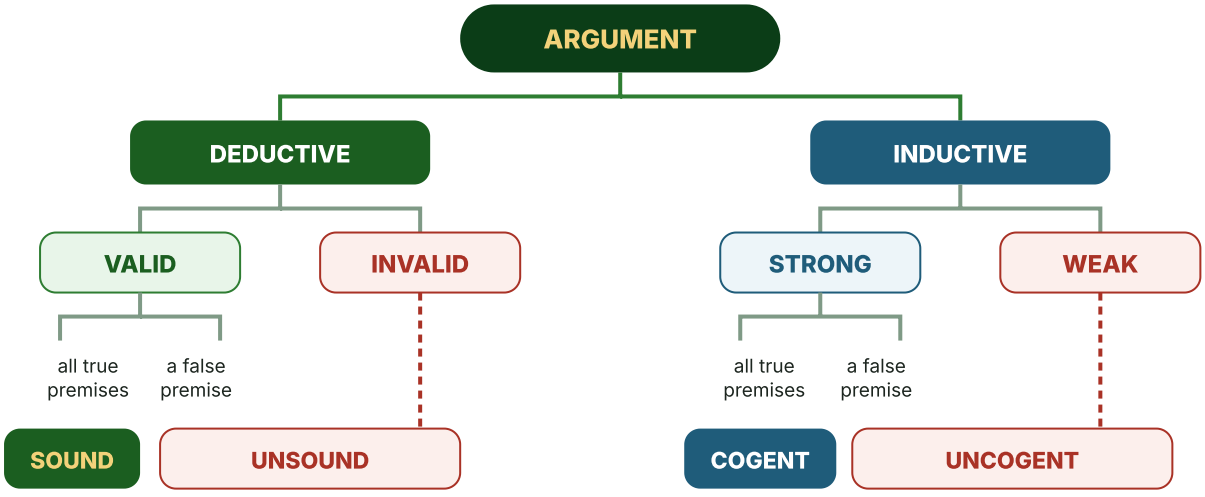
2.4.1 Two questions, in the right order

Every argument makes two claims: a **factual claim** (the premises present real evidence) and an **inferential claim** (the evidence supports the conclusion). The **inferential claim is the more important**: if the reasoning is bad, the argument is worthless. So we always:

★ **THE ORDER OF EVALUATION**

1st: Do the premises support the conclusion? (Assume the premises are true and test the link.) → valid / invalid · strong / weak

2nd: Are the premises actually true? (Only if the first test is passed.) → sound / unsound · cogent / uncogent



Invalid → always unsound · Weak → always uncogent

The complete evaluation map

2.4.2 Deductive arguments: validity

D **VALID AND INVALID**

Valid: if the premises are **assumed true**, it is **impossible** for the conclusion to be false. The conclusion follows with strict necessity.

Invalid: if the premises are assumed true, it is **possible** for the conclusion to be false, even though the conclusion is claimed to follow necessarily.

VALID

P1	All men are mammals.
P2	All bulls are men.
C	Therefore, all bulls are mammals.

INVALID

P1	All philosophers are rational.
P2	Socrates was rational.
C	Therefore, Socrates was a philosopher.

Two consequences follow:

1. **There is no middle ground.** An argument is **either valid or invalid**; validity has **no degrees**.
2. **Validity is only indirectly related to truth.** You do **not** need to know whether the premises are actually true to test validity. Assume they are true and ask: *could the conclusion still be false?*

Validity and truth values: the four combinations

Premises	Conclusion	Validity	Module example
True	True	**Valid or invalid**	Valid: all women are mammals... · Invalid: all philosophers are critical thinkers; Plato was a critical thinker; so Plato was a philosopher
True	**False**	**Invalid only**	All biologists are scientists; John Nash was a scientist; so John Nash was a biologist
False	True	**Valid or invalid**	Valid: all birds are mammals; all women are birds; so all women are mammals
False	False	**Valid or invalid**	Valid: all Americans are Ethiopians; all Egyptians are Americans; so all Egyptians are Ethiopians

★ **THE SINGLE MOST IMPORTANT POINT IN DEDUCTIVE LOGIC**

Any deductive argument with actually true premises and an actually false conclusion is invalid. This is the only combination that decides validity. The whole system of logic would be useless if it allowed us to start from truth and arrive at falsity.

! **EXAM FAVOURITES**

- A valid argument **can** have false premises and a false conclusion.
- An invalid argument **can** have all true premises and a true conclusion.
- If a valid argument has a false conclusion, **at least one premise must be false**.

2.4.3 Deductive arguments: soundness

D **SOUND**

A **sound argument** is a deductive argument that is **valid AND has all true premises**. If either condition is missing, it is **unsound**.

M THE TWO FORMULAS

SOUND = VALID + ALL TRUE PREMISES
COGENT = STRONG + ALL TRUE PREMISES

- Every **invalid** argument is unsound, whatever its premises.
- An unsound argument is **valid with a false premise**, or **invalid**, or **both**.
- Every sound argument has a **true conclusion** (true premises + validity guarantee it).
- A sound argument is what we mean by a **"good" deductive argument** in the fullest sense.

2.4.4 Inductive arguments: strength

D STRONG AND WEAK

Strong: if the premises are **assumed true**, it is **improbable** for the conclusion to be false.
Weak: if the premises are assumed true, it is **probable** for the conclusion to be false (the conclusion does not follow probably, though it is claimed to).

E THE APPLE BARREL

Strong: This barrel contains 100 apples. **Eighty** apples selected at random were tasty. Therefore, probably all 100 apples are tasty.
Weak: This barrel contains 100 apples. **Three** apples selected at random were tasty. Therefore, probably all 100 apples are tasty.

Two consequences, parallel to deduction, but with one big difference:

1. **Strength comes in degrees.** To be strong, the conclusion must be **more probable than improbable**. No inductive argument is absolutely strong or absolutely weak. A bigger sample or an extra premise can **strengthen or weaken** it.
2. **Strength is only indirectly related to truth.** To test it, assume the premises are true and ask whether the conclusion would **probably** be true.

Premises	Conclusion	Strength
True	True (probably)	Strong or weak
True	**False** (probably)	**Weak only**
False	True	Strong or weak
False	False	Strong or weak

2.4.5 Inductive arguments: cogency

D COGENT

A **cogent argument** is an inductive argument that is **strong AND has all true premises**. If either is missing, it is **uncogent**. Every weak argument is uncogent. The conclusion of a cogent argument is **probably true**.

! SOUND VS COGENT: ONE EXTRA REQUIREMENT

In a **sound** argument, true premises are enough. In a **cogent** argument, the premises must also **not ignore an important piece of evidence** that outweighs the given evidence and requires a different conclusion. (Ignoring such evidence is the fallacy of **suppressed evidence**, Chapter 5.)

	Deductive	Inductive
Claim	Conclusion follows **necessarily**	Conclusion follows **probably**
Good reasoning	**Valid** (no degrees)	**Strong** (has degrees)
Good reasoning + all true premises	**Sound**	**Cogent** (+ no suppressed evidence)
Conclusion of the good one	**Certainly** true	**Probably** true
Can contain formal fallacies?	Yes	No (only informal)

Exam Practice

28 QUESTIONS · LESSON 2.4

1 Jigjiga University · 2012

In a given argument, the factual claim is less important than the inferential claim.

True / False

2 Jigjiga University · 2012

For a deductive argument to be sound, it is enough to have all true premises.

True / False

3 Salale University · Mid 2015

In a valid deductive argument with a false conclusion, at least one premise must be false.

True / False

4 Salale University · Mid 2015

If an argument has all true premises and a true conclusion, it must be valid (if deductive) or strong (if inductive).

True / False

5 Salale University · Mid 2015

A strong inductive argument can have all true premises and a false conclusion.

True / False

6 Oda Bultum University · Mid 2017 E.C.

The truth and falsity of the premises and conclusion are necessary conditions for an argument to be valid.

True / False

7 Jigjiga University · Mid 2016

Identify the WRONG statement about good and bad arguments.

A) An argument can be well-supported (valid) even if its premises and conclusion are false
B) An argument can be bad even if its premises and conclusion are true
C) A bad argument is still an argument
D) An invalid argument can be a good argument

8 Jigjiga University · Mid 2016

Identify the odd (wrong) logical statement.

A) If all premises are assumed true, it is impossible for the conclusion of a valid argument to be false
B) In a valid argument the conclusion follows with strict necessity
C) If the premises do in fact support the conclusion, the argument is invalid
D) An invalid argument can have true premises and a false conclusion

9 Jigjiga University · 2012

Which one is FALSE?

A) Some unsound arguments are valid
B) Not all cogent inductive arguments are strong
C) Many deductive arguments are invalid or unsound
D) A strong argument with a false premise is uncogent

10 Jigjiga University · 2012

A strong inductive argument CANNOT have which combination?

A) True premises and probably true conclusion
B) True premises and probably false conclusion
C) False premises and true conclusion
D) False premises and false conclusion

11 Jigjiga University · 2012

Which is NOT always decisive in evaluating arguments?

A) The inferential link
B) The reasonableness of the inference
C) The support and relevance of the premises
D) The actual truth or falsity of the premises and conclusion

12 Jigjiga University · Mid 2014

Which of the following is FALSE about a valid deductive argument?

A) It is possible for the conclusion to be false if the premises are true
B) To test it, we assume the premises are true
C) It can have false premises
D) It has no middle ground with invalidity

13 Salale University · Mid 2015

Which statement about validity and truth is correct?

A) A valid argument must have true premises
B) An invalid argument cannot have true premises and a true conclusion
C) A valid argument can have false premises and a false conclusion
D) A sound argument can have a false conclusion

14 Salale University · Mid 2015

"All mammals are warm-blooded. All whales are mammals. Therefore, all whales are warm-blooded." This argument is:

A) Unsound
B) Sound
C) Invalid
D) Weak but cogent

15 Salale University · Mid 2015

"Every time I wear my lucky socks, my team wins. Today I am wearing them. Therefore, my team will win today." This is best evaluated as:

A) Deductive, valid, unsound
B) Inductive, strong, cogent
C) Inductive, weak (false cause)
D) Deductive, invalid (affirming the consequent)

16 Oda Bultum University · Mid 2017 E.C.

If an argument is valid but not sound, what can be said about it?

A) The premises are true and the conclusion is true
B) At least one premise is false
C) The premises are true but the conclusion is false
D) All are correct

17 Oda Bultum University · Mid 2017 E.C.

When evaluating a deductive argument, if all premises are true and the conclusion is false, the argument is:

A) Valid but not sound
B) Sound but not valid
C) Both valid and sound
D) Neither valid nor sound

18 Oda Bultum University · Mid 2017 E.C.

"Unlike monkeys, today's humans have feet that are not suited for grasping objects. Therefore, a thousand years from now, probably humans will still have such feet." This argument is:

A) Uncogent inductive
B) Invalid deductive
C) Cogent inductive
D) Weak inductive

19 Oda Bultum University · Exam 3

"Since Christmas is always on a Thursday, it follows that the day after Christmas is always a Friday."

A) Deductive, valid, unsound
B) Inductive, weak, uncogent
C) Deductive, invalid, unsound
D) Inductive, strong, cogent

20 Oda Bultum University · Exam 3

"This cash register drawer contains over 100 coins. Three coins selected at random were found to have dates earlier than 1945. Therefore, probably all of the coins in the drawer have dates earlier than 1945."

A) Inductive, strong, uncogent
B) Inductive, weak, uncogent
C) Deductive, valid, unsound
D) Inductive, strong, cogent

21 Oda Bultum University · Exam 3

"Since $x + y = 10$, and $x = 7$, it follows that $y = 4$."

A) Inductive, strong, uncogent
B) Deductive, valid, unsound
C) Deductive, invalid, unsound
D) Inductive, strong, cogent

22 Oda Bultum University · Mid 2017 E.C.

Evaluate: "Either Johar Mohammed is in Kenya or he is in the Netherlands. Johar Mohammed is in the Netherlands. Therefore, he is not in Kenya."

23 Oda Bultum University · Mid 2017 E.C.

Evaluate: "If it rained last night, then the streets are wet. It did not rain last night. Thus, the streets are not wet."

24 Hurley · Exercise 1.4

"Since Moby Dick was written by Shakespeare, and Moby Dick is a science fiction novel, it follows that Shakespeare wrote a science fiction novel."

A) Valid, sound
B) Valid, unsound
C) Invalid, unsound
D) Strong, uncogent

25 Hurley · Exercise 1.4

"All leopards with lungs are carnivores. Therefore, all leopards are carnivores."

A) Valid, sound
B) Invalid, unsound
C) Valid, unsound
D) Strong, cogent

26 Hurley · Exercise 1.4

"Every province in Canada has exactly one city as its capital. Therefore, since there are thirty provinces in Canada, there are thirty provincial capitals."

A) Valid, sound
B) Valid, unsound
C) Invalid, unsound
D) Inductive, strong

27 Hurley · Exercise 1.4

"People have been listening to rock and roll music for over a hundred years. Probably people will still be listening to it a year from now."

A) Strong, cogent
B) Weak, uncogent
C) Strong, uncogent
D) Valid, unsound

28 Hurley · Exercise 1.4

"Franklin Delano Roosevelt said that we have nothing to fear but fear itself. Therefore, women have no reason to fear serial rapists."

A) Strong, cogent
B) Weak, uncogent
C) Valid, sound
D) Strong, uncogent

Answers & Explanations

1. True

"The most important of the two is the inferential claim, because if the premises fail to support the conclusion, an argument is worthless."

2. False

Sound = **valid** + all true premises. True premises alone are not enough.

3. True

A valid argument can never have all true premises and a false conclusion.

4. False

True premises with a true conclusion allow both. "All philosophers are critical thinkers; Plato was a critical thinker; so Plato was a philosopher" is entirely true yet invalid.

5. False

Any inductive argument with actually true premises and an actually false conclusion is weak.

6. False

Validity depends on the link between premises and conclusion, not on their actual truth values.

7. D

An invalid argument can be a good argument

An invalid argument's premises do not support its conclusion, so it is always bad, and all invalid arguments are unsound. A is true ("All Americans are Ethiopians..." is valid), and B and C are true.

8. C

If the premises do in fact support the conclusion, the argument is invalid

The module says: if the premises do in fact support the conclusion, the argument is **valid**.

9. B

Not all cogent inductive arguments are strong

Cogent = strong + all true premises, so **every** cogent argument is strong.

10. B

True premises and probably false conclusion

True premises with a false conclusion means the support failed, so the argument is weak.

11. D

The actual truth or falsity of the premises and conclusion

Validity and strength depend on the relationship between premises and conclusion. Only one truth combination (true premises, false conclusion) is decisive.

12. A

It is possible for the conclusion to be false if the premises are true

In a valid argument, if the premises are true it is **impossible** for the conclusion to be false.

13. C

A valid argument can have false premises and a false conclusion

Example: "All Americans are Ethiopians; all Egyptians are Americans; so all Egyptians are Ethiopians."

14. B

Sound

Valid and all premises true, therefore sound. "Weak" and "cogent" belong to inductive arguments.

15. C

Inductive, weak (false cause)

It predicts the future from coincidences, so it is inductive, and it assumes an imaginary causal link (post hoc), so it is weak. (Some students argue for A; read as the module reads predictions and causal claims, C is intended.)

16. B

At least one premise is false

Valid + unsound means at least one premise is false. C is impossible for a valid argument. (The original option B read "the premises are false and the conclusion is true", one possible case.)

17. D

Neither valid nor sound

True premises with a false conclusion is always invalid, and an invalid argument cannot be sound.

18. C

Cogent inductive

It is a prediction (inductive), strongly supported, with a true premise.

19. A

Deductive, valid, unsound

The conclusion follows necessarily, but the premise is false (Christmas falls on different days).

20. B

Inductive, weak, uncogent

Three coins out of 100+ is far too small a sample, so the argument is weak, and every weak argument is uncogent.

21. C

Deductive, invalid, unsound

It is mathematical, so deductive. But y must be 3, so the conclusion does not follow: invalid, and therefore unsound.

22. **Deductive, invalid, unsound**

An "either ... or" premise does not rule out both being true, so affirming one disjunct does not prove the other is false.

23. **Deductive, invalid, unsound**

This is **denying the antecedent**: the streets could be wet for another reason (a burst pipe, cleaning).

24. B

Valid, unsound

If the premises were true, the conclusion would have to be true, so it is valid. But the premises are false (Melville wrote Moby Dick), so it is unsound.

25. B

Invalid, unsound

The premise speaks only of leopards **with lungs**, so the conclusion about **all** leopards does not follow necessarily. Hurley: invalid even though the premise and conclusion are true.

26. B

Valid, unsound

The arithmetic follows necessarily (valid), but Canada has ten provinces, not thirty, so a premise is false.

27. C

Strong, uncogent

If the premise were true, the conclusion would be probable, so it is strong. But rock and roll is not a hundred years old, so the premise is false: uncogent.

28. B

Weak, uncogent

The premise is true, but it gives no real support to this conclusion, so the argument is weak and therefore uncogent.

CHAPTER 2 · QUICK REVISION

Basic Concepts of Logic — at a glance

Argument or not? The quick test

If you see...	It is probably...
Indicator word + a real inferential link	**Argument**
"Thus" + examples showing what or how	**Illustration** (non-argument)
"Because" + an accepted fact being explained	**Explanation** (non-argument)
A single "if ... then ..."	**Conditional statement** (non-argument)
A topic sentence developed, nothing doubtful proved	**Expository passage** (non-argument)
"Since" meaning *from the time that*	**Not** an argument

Deduction vs induction at a glance

Feature	Deductive	Inductive
Strength of claim	Necessarily (impossible to be false)	Probably (improbable to be false)
Indicator words	certainly, necessarily, absolutely, definitely	probably, likely, plausible, reasonable to conclude
Typical forms	Mathematics, definition, categorical / hypothetical / disjunctive syllogism	Prediction, analogy, generalization, authority, signs, causal inference
Evaluation	Valid/invalid → sound/unsound	Strong/weak → cogent/uncogent
Degrees?	No (either valid or invalid)	Yes

! THE TEN STATEMENTS EXAMINERS LOVE

1. A bad argument is still an argument.
2. A single conditional is not an argument.
3. An indicator word does not guarantee an argument.
4. Deduction is **not** "general to particular".
5. Statistics → inductive; geometry → deductive.
6. True premises + false conclusion → **invalid / weak**, always.
7. A valid argument can have false premises and a false conclusion.
8. Sound = valid + all true premises; cogent = strong + all true premises.
9. Invalid → unsound; weak → uncogent.
10. In "if A then B", A is **sufficient** and B is **necessary**.